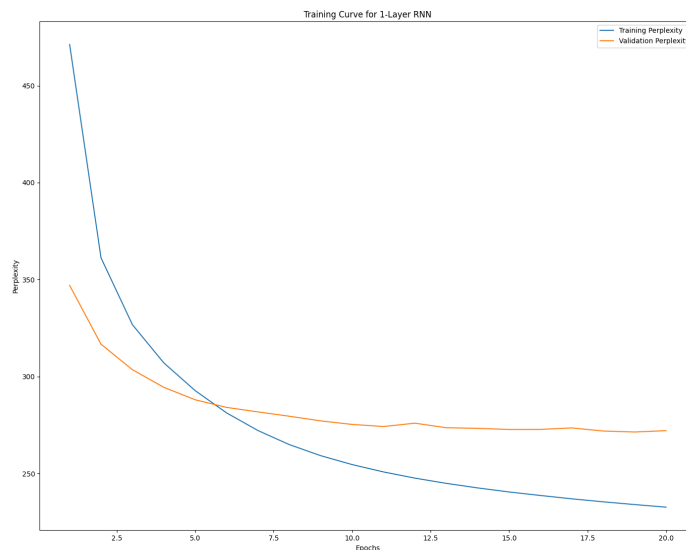


Deep Learning Homework 3

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1. The Vanilla RNN:

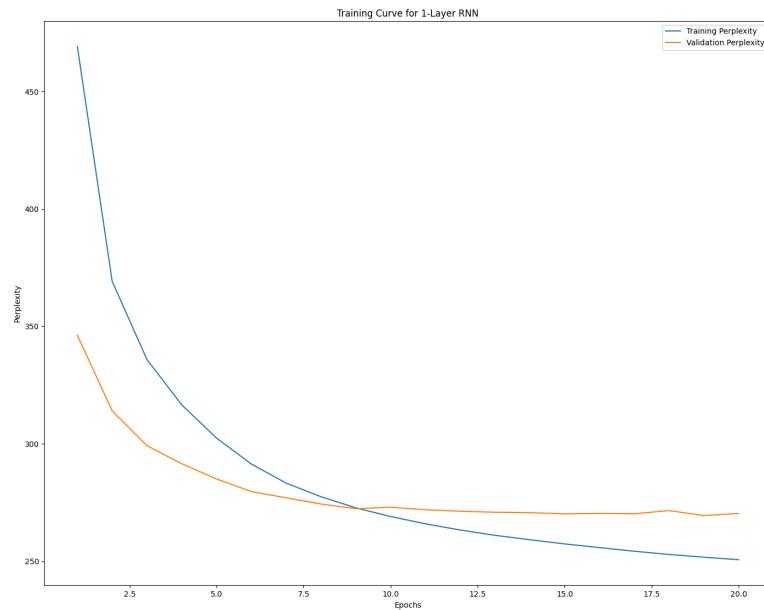
- a. We implemented a vanilla RNN using a vocabulary size of 9892, mapping any tokens outside of this vocabulary to `<unk>`. This is the number of tokens that appear more than 16 times in the training corpus. Input tokens were first passed to a *torch Embeddings* layer of dimension 100. These embeddings were then passed into a single-layer RNN with a hidden dimension of 30 and a window size of 10, and the final hidden layer output was passed through a 10x9892 feed-forward layer. We also clip the gradients to 2 within each batch to stabilize training. We used cross entropy loss as a performance metric for training, and reported perplexity as an evaluation metric.
- b. The model was trained over 20 epochs with the Adam Optimizer using the default learning rate of 0.001. The window size and the hidden dimension were tuned after substantial trial and error to achieve the desired test perplexity. We noticed through this process that (strangely) the validation perplexity was noticeably higher than the test perplexity. It is possible this is simply due to the test set involving “easier to predict” sentences relative to the validation set. Note we end with a validation perplexity of about 318, but achieve a noticeably better test perplexity. As requested, the model is approaching training perplexity convergence at the 20 epoch mark, while improvements in the validation perplexity stop significantly improving after the 5th epoch.
- c. Below are the associated training and validation curves.



- d. We achieved a final test perplexity 248.6.

2. The improved RNN:

- a. We chose to improve the architecture of the vanilla RNN by implementing a second layer with a dropout of 0.1. This allows the model to more easily capture higher-order patterns in token sequences, and adding the associated dropout prevents overfitting with the increase in the number of neurons. Keeping the `hidden_size` at 30, and the window size at 10, we saw similar but slightly improved performance. Below are the associated training and validation curves.



We achieved a final test perplexity of 247.0.